Willie Rush Lim

Employment

Brown University, RI	Dean's Tamarkin Assistant Professor	2026 – now
	Tamarkin Assistant Professor	2024 – 2026

Education

Stony Brook University, NY	PhD in Mathematics ○ Dissertation: "*From Herman Rings to Herman Curves*" ○ Advisor: Dzmitry Dudko	2019 – 2024
Imperial College London, UK	MSci in Mathematics	2015 – 2019

Research Interests

Complex dynamics and renormalization theory, at the interface of analysis and geometry

Publications

[3] Hyperbolicity of renormalization of critical quasicircle maps
Comm. Math. Phys. 407 (2026): 88. (94 pp)

[2] Rigidity of J-rotational rational maps and critical quasicircle maps
Trans. Amer. Math. Soc. 379 (2026): 2507–2567.

[1] A priori bounds and degeneration of Herman rings with bounded type rotation number
Invent. Math. 242 (2025): 827–893.

Preprints

[8] Wandering continua, renormalization, and Lebesgue measure on Julia sets
arXiv:2611.05949 (11 pp)

[7] Rigidity of the attractor of neutral renormalization (with D. Dudko and M. Lyubich)
arXiv:2609.37383 (118 pp)

[6] Lebesgue measure of the postcritical set of neutral quadratic polynomials
arXiv:2609.35563 (46 pp)

[5] Uniform bounds for bubbles of neutral quadratic polynomials (with D. Dudko)
arXiv:2609.15500 (33 pp)

[4] The combinatorics of sector renormalization
arXiv:2607.11408 (39 pp)

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Research Visits

Isaac Newton Institute (INI) for Mathematical Sciences, Cambridge, UK — 2026 July
Invited participant for the Complex Dynamics: Interactions and Influences programme (4 weeks)

Mathematical Institute of the Polish Academy of Sciences, Warsaw, Poland — 2026 Apr – May
Invited early career researcher position for the Simons Semester: Continued Fractions, Fractals, Ergodic theory and Dynamics (5 weeks)

Invited Talks

Conferences and workshops:

Complex dynamics: a century on from Fatou and Julia. INI, UK	2026 July
Simons Semester: Continued Fractions, Fractals, Ergodic Theory, and Dynamics. Banach Center, Poland (4.5 hours of mini-course)	2026 May
Fields-CNAM Nonlinear Days 2025. Fields Institute, Canada	2025 Aug
Geometric Complexity of Julia Sets VI. Bedlewo, Poland	2025 Aug
Topics in Complex Dynamics. University of Barcelona, Spain	2025 June
AMS Sectional Meeting. UW-Milwaukee	2024 Apr

Seminars:

Dynamical Systems Seminar, Stony Brook University	2022 Nov, 2024 Nov, 2026 June
Geometry and Topology Seminar at Brown and Yale (GATSBY)	2026 Apr
Arithmetic and Complex Dynamics Seminar, Brown University	2026 Apr
Geometry & Topology Seminar, Brown University	2024 Jan, 2025 Feb
Dynamical Systems Seminar, Cornell University	2024 Nov
Dynamical Systems Seminar, Boston University	2024 Oct
Complex Analysis and Dynamics Seminar, CUNY	2023 Mar

Teaching Experience

At *Brown University*:

Instructor	Math 1710 Topology	2025 Fall, 2026 Fall
	Math 1110 Ordinary Differential Equations	2025 Fall, 2026 Fall
	Math 1040 Fundamental Problems of Geometry	2026 Spring
Lecturer	Math 0180/0200 Multivariable Calculus	2024 Fall, 2025 Spring

At *Stony Brook University*:

Instructor	MAT127 Calculus C (for Simons STEM Scholars)	2023 Summer
	MAT203 Calculus III	2021 Summer, 2022 Summer
	MAT342 Applied Complex Analysis	2020 Summer
Lecturer	MAT125 Calculus A	2022 Fall
TA	Calculus I, II, IV, Intro. to Linear Algebra, Applied Real Analysis, Dynamics and Chaos, Graduate Complex Analysis	2019 – 2024

Honors and Awards

Dean's Tamarkin Assistant Professorship, Brown (department-wide research honor awarded to one recipient)	2026
Chairman's Award for Excellence in Teaching, Stony Brook	2024
IBM Prize for Excellence in Pure Mathematics	2019
Prize for Excellence in Support of Teaching, Imperial College	2019
G-Research Prize (awarded to exceptional non-final year students)	2017
President's Undergraduate Scholarship, Imperial College	2015 – 2019

Mentoring

Undergraduate honors thesis advisor, Brown (1 student, ongoing)	2026 – present
Instructor of the Mathematics Summer Program (MathCamp) for high school students	2022
Directed Reading Program (2 students, on wallpaper groups and fractal geometry)	2021 – 2022

Service

Referee for *J. Fractal Geom.*, *J. Differ. Equ. Appl.*, *Invent. Math*.	
Reviewer for *Mathematical Reviews* (MathSciNet)	
Co-organizer of Horizons Seminar, Brown	2026 – present
Committee member for an oral qualifying exam for a graduate student, Brown	2025
Co-organizer of ENYGMMa (Empowering New York Gender Minority Mathematicians)	2022
Co-organizer of the Graduate Student Seminar, Stony Brook	2020 – 2021